Thesis Description

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In recent years, the invasive Asian hornet (*Vespa velutina*) has appeared across Europe, posing a significant threat to local wildlife and apiaries. Multiple techniques have been deployed to contain this threat, including visual tracking, triangulation, harmonic radar and radio telemetry.

This thesis investigates radio telemetry to track Asian hornets. The primary focus is to design a cost-effective, rechargeable, sub-200 mg transmitter that is practical and user-friendly for non-technical operators. The user must be able to attach and activate the transmitter in a simple manner and track the hornet with a portable receiver. This thesis aims to improve upon existing wildlife trackers by being purpose-built specifically for the tracking and eradication of Asian hornets.

The transmitter must minimize energy consumption and component count. Weight minimization is a critical aspect of this research, offering key practical advantages: reduced tag weight allows a larger battery to be integrated to extend operational lifespan or increase RF output power, while enabling a higher proportion of hornets to carry the payload over greater distances.

Primary Objectives

- Literature review of existing (analog) transmitters.
- Build a theoretical model and simulation of an analog transmitter.
- Develop an initial prototype to verify the simulations.
- Optimize the transmitter for hornets (<200 mg weight and attachable to hornets).
- Test the transmitter range and frequency stability, fine-tune where needed.
- Demonstrate field tracking of hornets over distances up to 1000 m in real-world conditions.

Planning

